



# LUCIE: A Lightweight Uncoupled Climate Emulator

with long-term stability and physical consistency

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## Research summary

- LUCIE is the most lightweight data-driven climate emulator that requires only 1 A100 GPU and 2.4 hour training time
- It can generate over 6000 years of stable and physical prediction within 24 hrs.
- LUCIE can generate extreme values beyond the training domain.

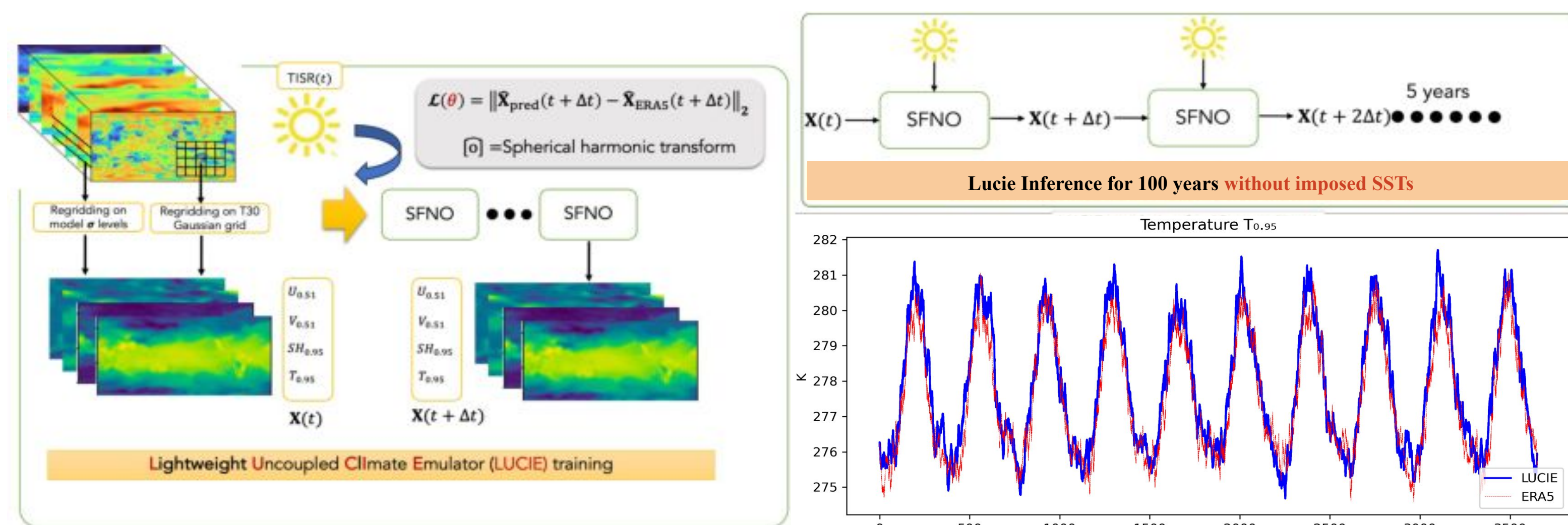
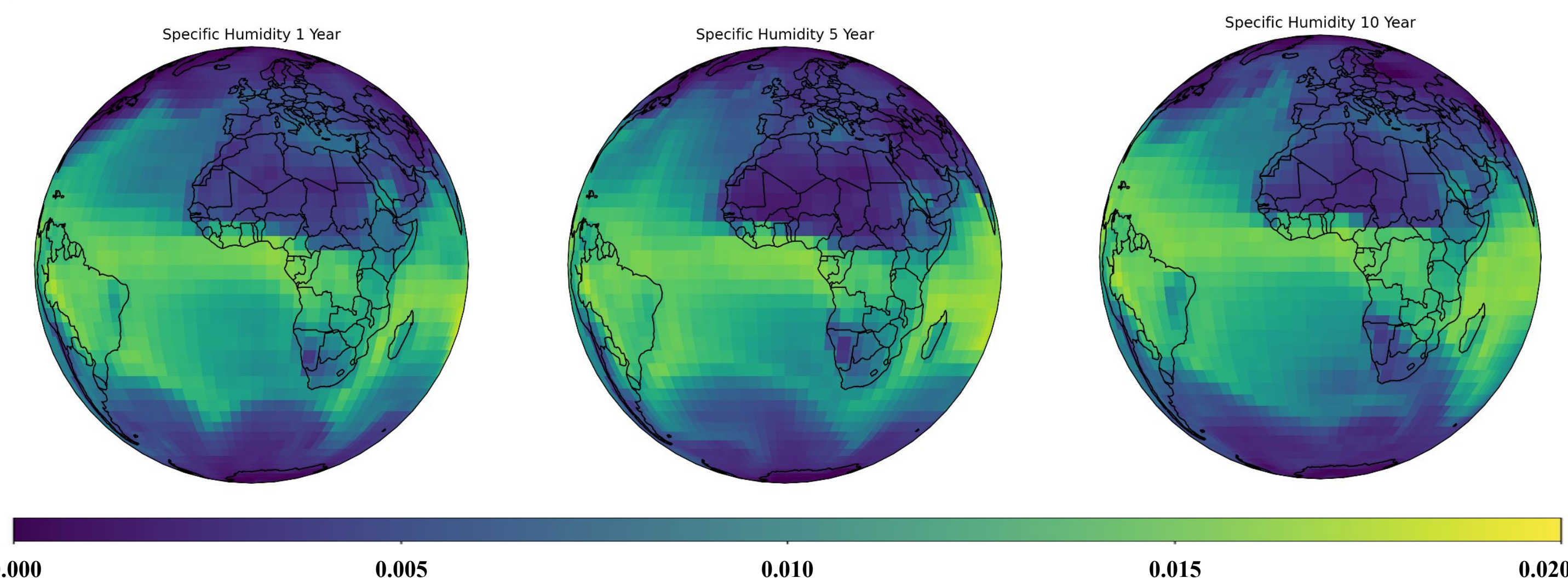


Fig: LUCIE schematics

| Model     | Device   | Time    | Model       | Device   | Time    |
|-----------|----------|---------|-------------|----------|---------|
| ACE       | 4 A100   | 63 hrs  | Stormer     | 128 A100 | 24 hrs  |
| GraphCast | 32 TPU   | 4 weeks | FourCastNet | 24 A100  | N/A     |
| ClimaX    | 80 V100  | N/A     | FuXi        | 8 A100   | 1 week  |
| Pangu     | 192 V100 | 8 weeks | LUCIE       | 1 A100   | 2.4 hrs |

Fig: Computing resources requirement

## 10 year inference



## Climatology Bias

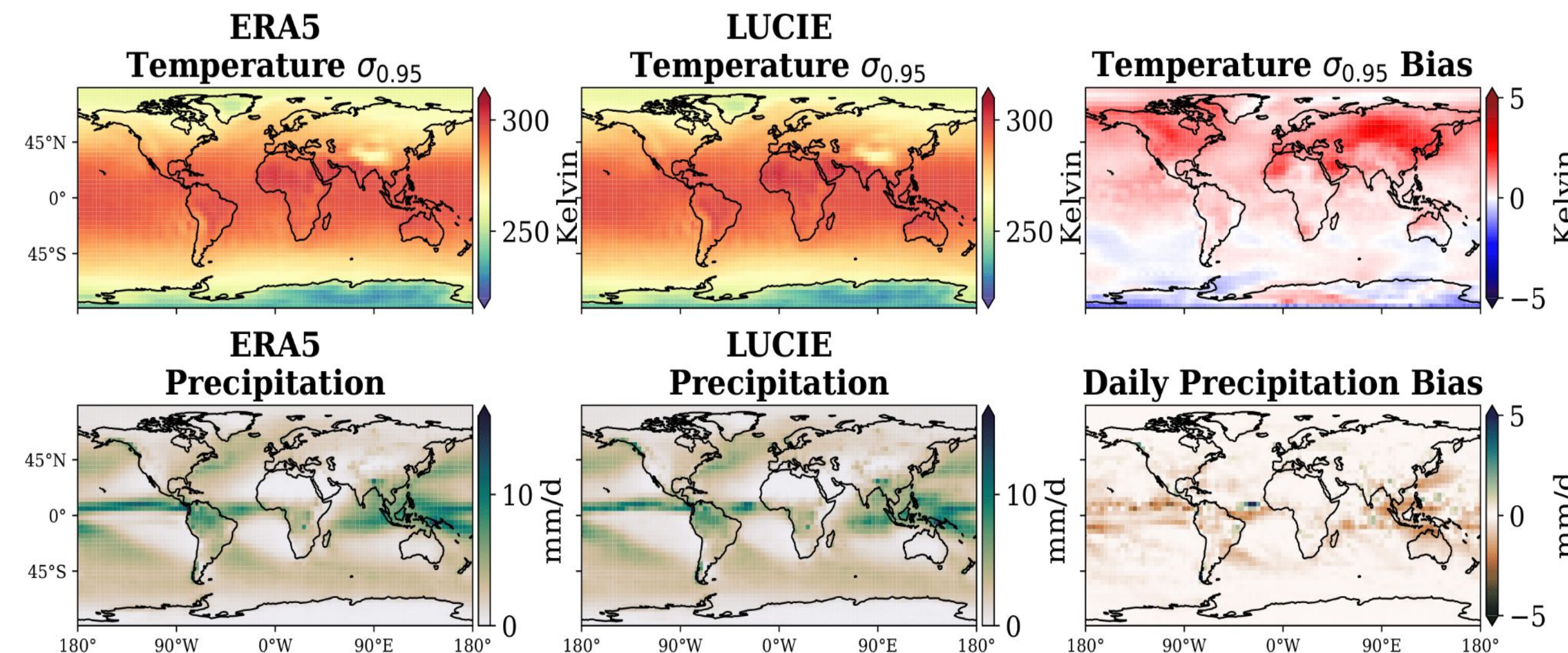


Fig: Climatology map

## Extreme events

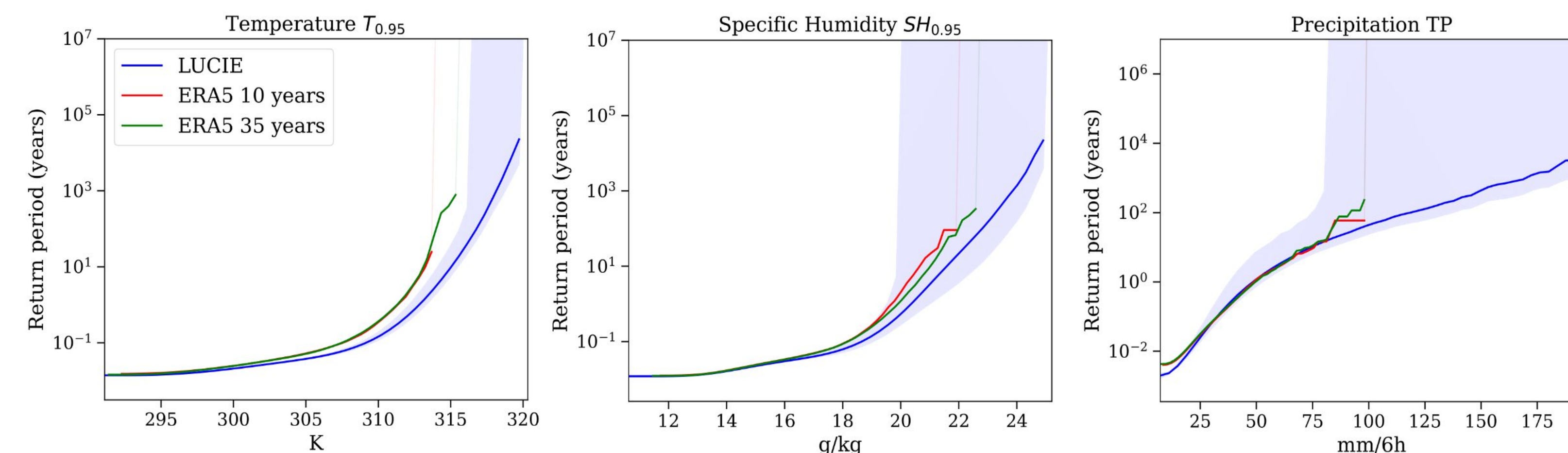
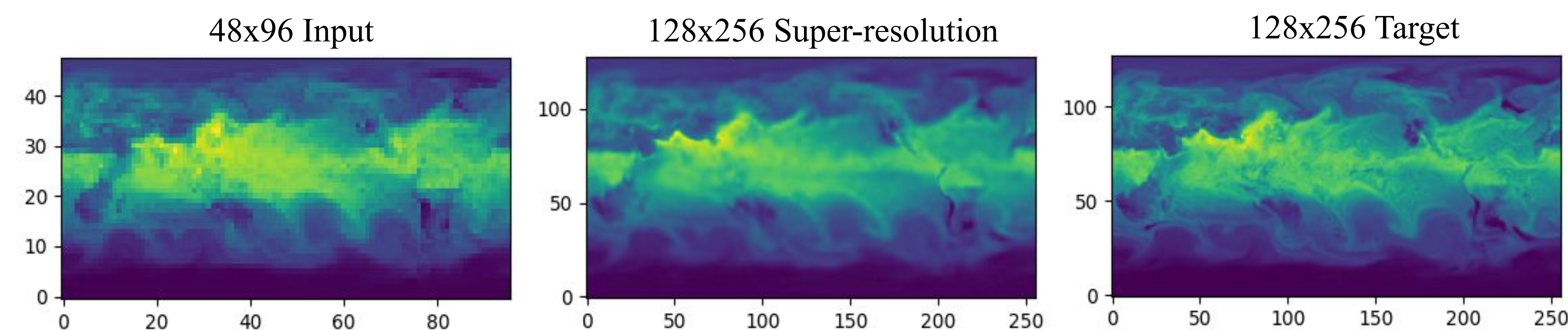


Fig: Return period of ERA5 10 years, 35 years, and LUCIE

## Super-resolution



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Preprint : H. Guan, T. Arcomano, A. Chattopadhyay, and R. Maulik. "LUCIE: A Lightweight Uncoupled Climate Emulator with long-term stability and physical consistency for O(1000)-member ensembles" arXiv preprint arXiv:2405.16297 (2024).